

JARED ASHER MANANGUIT

BS Electronics Engineering (ECE) • University of Santo Tomas • 3rd Year
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EDUCATION

University of Santo Tomas — Faculty of Engineering

2023 – Present

BS Electronics Engineering (ECE)

- Dean's Lister, Academic Year 2023–2024
- DOST-SEI Scholar

Relevant Coursework: Computer Networks 1 & 2, Microcomputer Systems and Design, Switching Circuits and Logic Design, Digital Communications, Communication Systems Theory, Signal Processing Theory and Applications, Control Systems Theory

TECHNICAL SKILLS

Programming: Python, C++, Verilog

Simulation & Design: MATLAB / Simulink, LTSpice, KiCad, AutoCAD, Quartus II

Embedded & Hardware: Arduino, PlatformIO, PCB design

PROJECTS

Undergraduate Thesis — Multi-Agent Reinforcement Learning for Bus Scheduling

In Progress

- Co-writing a research review on multi-agent reinforcement learning (MARL) for traffic scheduling, as part of a four-person team.
- Studying why MARL models that work in simulation struggle in real traffic, and where the current research falls short.

Game Show Buzzer System — Embedded Software

In Progress

- Writing the Arduino code for a real-time buzzer system, including the logic that keeps the LED displays and audio in sync.
- Managing the code and merging the team's work on GitHub, using VSCode and PlatformIO for testing.

Class AB Power Amplifier — Circuit Design & PCB Implementation

2025

- Designed a 25-Watt Class AB audio amplifier and laid out the PCB in KiCad.
- Used LTSpice to pick component values and check performance, then built and tested the actual board.

Communication System Design

2025

- Helped build a digital communication model in MATLAB / Simulink to test demodulation and QAM performance.

Arithmetic and Logic Unit (ALU) Design

2025

- Designed and simulated a multi-function ALU in Verilog (Quartus II), checking the logic with waveform simulations.

CERTIFICATION

- Cisco Certified Network Associate 1 (CCNA 1) — Cisco Networking Academy

LEADERSHIP & INVOLVEMENT

Engineering Student Council — University of Santo Tomas

2024 – 2025

- Official photographer for council and college-wide engineering events; documented activities and kept the photo archives organized.